

Rushil Malode

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EDUCATION

University of California, San Diego

Fall'26 - Spring'28

Master of Science in Electrical and Computer Engineering, Machine Learning and Data Science

Indian Institute of Technology, Bombay

Fall'22 - Spring'26

B.Tech. in Electrical Engineering with a Minor in Artificial Intelligence and Data Science

CPI:8.29/10.00

PROFESSIONAL AND RESEARCH EXPERIENCE

Robust Embedding Selection for Canonical Data Mapping

Saket Lab, IIT Bombay

A Research Project under Prof. Saket Choudhary, Koita Centre for Digital Health (KCDH)

Present

- Developed a framework to select embedding/decoder models for mapping corrupted text entries to canonical keys
- Modeled embedding robustness by injecting directional Gaussian noise on the **von Mises-Fisher (vMF) distribution**
- Engineered a dimension-scaled adjustment to standardize noise concentrations across embedding dimensions
- Investigating theoretical guarantees on retrieval accuracy under the modeled embedding perturbations

Interpretable Machine Learning for Credit Risk Scoring

ICICI Bank

Internship Project on Credit Risk Scoring under the Risk Team

Summer'25

- Implemented classification models for risk based borrower segmentation to enhance loan approval decisions
- Analyzed and processed **2.2M+** loan records and **150+** borrower features to uncover insights on borrower behaviour
- Achieved **0.68 ROC-AUC** on training **7** neural and tree-based ML models including **XGBoost** and **LightGBM**
- Implemented **Shapely Additive Explanations** and **Counterfactual Explanations** to enhance transparency

Best Arm Identification in Correlated Grouped Bandits

Stochastic Systems Lab, IIT Bombay

Bachelor's Thesis Project under Prof. Sharayu Moharir, Department of Electrical Engineering

Fall'25

- Developing a novel best-arm-identification algorithm to exploit reward correlation in Grouped Multi-Armed Bandits
- Performed an extensive literature review including over **37** publications in the **Multi-Armed Bandit** domain
- Generated **10M+** correlated rewards via parallelization using **rpy2** and **BINCOR** to evaluate empirical correctness
- Achieved **50%** lower complexity compared to the existing Grouped Bandit on MovieLens and Goodreads datasets

Network Inversion on Convolutional Neural Networks

MeDAL Lab, IIT Bombay

A Research Project under Prof. Amit Sethi, Department of Electrical Engineering

Fall'25

- Improved upon a novel generator-classifier architecture to **reconstruct training-like data** from pre-trained classifiers
- Reconstructed distinct training-like dataset samples for MNIST, FMNIST and CIFAR from classifiers trained on **10, 50 and 100 images**, using classifier properties including gradient magnitudes, cross entropy and KL-divergence
- Worked towards improving reconstruction diversity and generalizing over standard neural network architectures

CASI-3D U-Net for ¹³CO Bubble Prediction

Max Planck Institute for Radio Astronomy

A Research Project under Prof. Dario Colombo and Kartik Neralwar

Summer'24

- Utilized CASI-3D U-Neural Net on processed data-cubes to detect **¹³CO stellar bubbles** and validated their position
- Conducted **data exploration** on SEDIGISM data for **PPV cubes** and identified **velocity channels** with signals
- Post-processed pixel feedback probabilities by **setting precise cutoffs** for enhanced **bubble-edge detection**
- Validated prediction positions to WISE Catalog of Galactic **HII Region** to ensure positional accuracy of predictions

Streaming Model for Volume of Union of Convex Bodies

TU Wien

A Research Project under Dr. Sagar Malhotra, TU Wien Informatics

Summer'24

- Ideated on an innovative randomized streaming algorithm for volume of union of convex bodies under space constraints
- Performed in-depth analysis of the **Karp-Luby-Madras (KLM)** randomised approximation for union of sets
- Understood KLM in **Approximating Volume of Unions** and **Approximate Weighted Model Integration**
- Developed an algorithm for the setting by devising a novel density based sampling logic and extending the size estimation streaming algorithm from **Estimating Size of the Union of Sets in Streaming Model** for volumes

KEY ACADEMIC PROJECTS

Marked System-2 Recommenders

CS 6103: Human Centred AI

Guide: Prof. Arpit Agarwal, Department of Computer Science and Engineering

Spring'26

- Proposed a marked dual-kernel Hawkes process preserving identifiability beyond decay-rate collapse in recommenders
- Derived identifiability and MLE consistency guarantees for marked temporal point processes with engagement signals
- Developed a branching-EM inference framework, achieving **5.2×** lower estimation error and **98%** latent-state recovery

Image Captioning for Remote Sensing

EE 782: Advanced Topics in Machine Learning

Guide: Prof. Amit Sethi, Department of Electrical Engineering

Fall'25

- Developed **8** Transformer and LSTM based image captioning models and performed rigorous model evaluation
- Fine-tuned **ResNet-18** paired with LSTM and Transformer decoder on **10K+** satellite images from RSCID data
- Performed qualitative analysis and evaluated results using BLEU and METEOR scores with attention map visualization
- Compared **MobileNet** and **ResNet-18** backbones across memory efficiency, inference speed and performance metrics

Efficient Lightweight Skin Disease Prediction

DH302: Public Health Informatics

Guide: Prof. Saket Choudhary and Prof. Nirmal Punjabi, Koita Centre for Digital Health (KCHD)

Fall'25

- Developed lightweight and quick-inference skin lesion classifiers for resource constrained environments
- Conducted comprehensive literature review on efficient vision transformer architectures for medical image classification
- Performed data harmonization by mapping BCN-20000 dermatology dataset labels to HAM-10000 taxonomy
- Evaluated MobileViT and LOL-ViT architectures with modified attention mechanisms from Linformer and Nyströmformer

Probabilistic Inference on FHMMs and Turbo Codes

CS 791: Probabilistic Foundations of AI

Guide: Prof. Sunita Sarawagi, Department of Computer Science and Engineering

Fall'25

- Performed inference on Factorial Hidden Markov Models (FHMMs) and built decoders for noisy codes
- Implemented exact inference for Factorial-HMMs by graph triangulation and using junction trees with **message-passing**
- Developed **Loopy Belief Propagation** for FHMM marginals through iterative message passing on factor graphs
- Built **Viterbi**, **BCJR** and **Turbo** decoders for noisy RSC codes, reducing BER to near-zero on **1000-bit** sequences

Voice and LLM based AI Room Guard

EE 782: Advanced Topics in Machine Learning

Guide: Prof. Amit Sethi, Department of Electrical Engineering

Fall'25

- Built a robust face-detection pipeline using **DeepFace**, **VGG-Face** and **Random Forest** to guard a territory
- Leveraged **Gemini LLM** to setup a 4-tier generative escalation warning system with **gmail SMTP** alerts
- Implemented voice-command activation using **speech recognition** and **gTTS** for multi-modal interaction

FPGA-based Resistive RAM Tester

EE 344: Electronic Design Lab

Guide: Prof. Veeresh Deshpande, Department of Electrical Engineering

Spring'25

- Built an **FPGA-based** Resistive RAM tester integrating analog circuitry, FPGA logic, and a web-based user interface
- Designed analog front-end circuits including **DAC** and **ADC** drivers, **op-amps** on a custom **PCB** based on published RRAM research, and verified functionality with FPGA-generated **500 ns** pulses achieving **95% accuracy**
- Hosted web-interface on the Zynq-7000 processor of **PYNQ-Z2**, enabling real-time user-driven read/write operations

Language Translation using Neural Machine Translation

CS 626: Natural Language Processing

Guide: Prof. Pushpak Bhattacharyya, Department of Computer Science and Engineering

Fall'24

- Designed a language-to-language translation based on Transformer encoder-decoder using Bahdanau attention
- Implemented **Transformer** architecture from “Attention Is All You Need” in PyTorch for neural machine translation
- Built **Bahdanau attention** in encoder-decoder framework to improved alignment in language-to-language translation
- Developed a **Seq2Seq** machine translation pipeline in **PyTorch** to evaluate English-to-Hindi translation accuracy

RISC Microprocessor Design

EE 309: Microprocessors

Guide: Prof. Virendra Singh, Department of Electrical Engineering

Spring'24

- Designed a **16-bit RISC** processor featuring a **6-stage pipeline** architecture capable of executing **26 instructions**
- Implemented **hazard mitigation** techniques with **forwarding**, **flushing**, **stalling** and **branch prediction**
- Simulated individual microprocessor components such as **Instruction Memory**, **Pipeline Registers**, **Controller Boxes**, **Register File**, **ALU**, **MUXes**, and **Sign Extenders** using **VHDL** on Intel Quartus software

Practical Applications of Control Systems

EE 324: Control Systems Lab

Guide: Prof. Debraj Chakraborty, Department of Electrical Engineering

Fall'24

- **Line Follower Bot**: Devised a **PID controller** taking inputs from the **infrared sensors** to calculate the velocity
- **Inverted Pendulum**: Implemented a **Linear Quadratic Regulator** controller on **Arduino**, which takes the current pendulum **position** as the input and outputs the motor **torque** necessary to keep the pendulum **upright**
- **Noise Cancellation**: Designed a **second order filter circuit** using simple **Operational Amplifiers**, **resistors** and **capacitors**, and simulated it on **MATLAB** for verification, to **filter out** the noise in a box by **20 decibels**

Assembly & Embedded C with Intel 8051

EE 337: Microprocessor Lab

Guide: Prof. Nikhil Karamchandani, Department of Electrical Engineering

Spring'24

- Programmed **8051** based **Pt-51** micro-controller using **Assembly** and **Embedded C** for embedded systems
- Implemented **keyboard interfacing** using **timers** and **external interrupts** for stopwatch and **musical notes**
- Integrated **SPI** to interface an **analog-to-digital converter** (ADC) **MCP3008** with the 8051 micro-controller
- Established **serial communication** with a **USB-UART** module and executed a **low-pass digital FIR filter**

Digital Logic Design in VHDL

EE 214: Digital Circuits Lab

Instructor: Prof. Siddharth Tallur, Department of Electrical Engineering

Fall'23

- Gained proficiency in **VHDL** through experience in **Structural**, **Behavioral**, and **Dataflow** modeling techniques
- Designed and Implemented **10+ Components** on Intel Quartus including Sequence Generators and FSMs
- Verified the designed components by manual hardware testing and **scan-chain** on the **FPGA XEN-10** boards

ADDITIONAL PROJECTS

Quantum Computing with Qiskit

Maths and Physics Club, IIT Bombay

Winter'24

- Developed **SWAP**, modular arithmetic and entanglement circuits and visualized states on the **Bloch sphere**
- Implemented **Deutsch–Jozsa**, **Bernstein–Vazirani**, **GHZ game**, **QFT**, **QPE**, **Shor** and **Grover** algorithms
- Applied **density matrices** and **phase kickback** principles to design advanced quantum circuits in Qiskit

Reinforcement Learning on Atari Games

Summer of Code by Web and Coding Club, IIT Bombay

Summer'24

- Built ϵ -greedy bandit algorithm, **4×4 Gridworld agent**, and understood **Markov Decision Processes**
- Implemented taxi-environment using OpenAI Gym and compared reward curves for **Q-Learning** and **Monte Carlo**
- Designed **Deep Q Networks** and trained them to master **Cart-pole**, **Pong** and **Breakout** with a high win rate

Understanding Credit Score Algorithms

FinSearch Competition held by Finance Club, IIT Bombay

Summer'24

- Conducted an in-depth study of **credit scores**, **credit scoring models**, and the functioning of credit systems
- Analyzed the impact of credit scores and **credit cycles** on fintech **P2P**, **populace** and **business lending** firms
- Achieved over **80%** accuracy by building a default-based credit model on **10+** features using **Random Forest**

MENTORSHIP, LEADERSHIP AND TEACHING ROLES

Senior Student Consultant | ShARE, IIT Bombay

Fall'23 - Spring'26

Future Leaders Program with 2800+ students from over 100+ top universities across 41 countries

- Selected among **300+** applicants through a rigorous procedure based on analytical and communication skills
- Underwent **40+** hours of training by **ex-MBB** consultants on problem-solving and effective communication
- Worked as a technical consultant with Nutribinge Fitfire to achieve the right market entry and product-market fit
- Spearheaded teams in driving growth for Whizlabs, a Global Ed-Tech Start-up, and Digigrain, an Agri-tech startup

Department Academic Mentor | Electrical Engineering

Fall'24 - Spring'25

- Mentoring **12 Sophomores** on a one-to-one basis and inspiring them for academic and co-curricular endeavours
- Part of a 54 member team, selected out of **170+ applicants** after rigorous peer reviews and interview stages

Teaching Assistant | ELIT, Institute Student Mentorship Program

Fall'25

- Part of a **13+** member team to enhance students' fluency in English and strengthen their **communication skills**
- Curated English **learning resources** for soft skills, grammar and writing to coach **50+** students in the course

TECHNICAL SKILLS

- **Languages:** Python, C++, C, VHDL, Embedded C, Assembly, HTML
- **Libraries:** PyTorch, TensorFlow, Keras, NumPy, SciPy, Pandas, Scikit-learn, Matplotlib, OpenCV, Astropy, NLTK
- **Software:** AutoDesk Fusion 360, Quartus Prime, KiCAD, GNU Radio, Git, Keil, LTSpice, NGSpice, L^AT_EX

KEY COURSES UNDERTAKEN

Machine Learning	Human-Centred AI, Statistical Learning Theory, Advanced Topics in Machine Learning, Online Learning and Optimization, Game Theory and Algorithmic Mechanism Design, Natural Language Processing, Programming for Data Science, Markov Chains and Queuing Systems, Probability and Random Processes, Signal Processing, Linear Algebra, Calculus
Electrical Engineering	Communication Networks, Control Systems, Electromagnetic Waves, Electronic Devices & Circuits, Microprocessors, Analog Circuit, Digital Systems, Power Engineering
Others	Teaching Assistant Skill Enhancement & Training, Economics, Introduction to Entrepreneurship, Operations Analysis, Introduction to Bio-nanotechnology

EXTRACURRICULARS

Filmmaking	Assistant Director and Script-writer of gold winning films at Inter-IIT Culturals Filmed 6 short films alongside interdisciplinary teams of 15+ crew members
Awards	Recipient of the all rounder award at Sarala Birla Academy for the year 2018-2019 Participated in the Guinness World Record for Asia's largest taekwondo display
Social	Volunteered at 2 Round Square Conferences , 200+ international participants each
Oratory and Debate	Speaker at Josh Talks , a social tech startup, reaching an audience of 12K+ viewers Participant at 4 prestigious Model United Nations and winner of 2 verbal mentions
Sports	FIDE Rated chess player rated 1455 (std.) and a bronze medalist at the state-level